Pricing the Impermanent:
An Option-Implied Framework for Hedging UniV3 LP Loss

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Abstract. Topaz Blue and Bancor (2021) showed that, in aggregate, Uniswap v3 liquidity providers lost more to impermanent loss (IL) than they earned in fees — an empirical, backward-looking diagnosis. We extend it 4 ways:

- An independent **position-level re-measurement** across 5 major ETH pools confirms the result — **$50.7m of IL across 27,529 closed positions, every pool negative** — using a cleaner estimator built on NonfungiblePositionManager token-IDs, valued at each position's exact exit price.
- We formalise IL as a **short-gamma options position** and price the protection off a static snapshot of the Deribit volatility surface via SABR-calibrated static replication, treating the capped 'Limited' product as a **first-passage barrier** rather than a European condor.
- We quantify the rebalancing **'gamma bleed'** the original authors flagged as future work, with a Monte-Carlo calibrated to realised volatility, fee APR, and TVL computed from **Dune Analytics**.
- We close with 3 **worked examples** — representative positions from the measured cohort — showing step-by-step how an LP would have priced, bought, and settled the hedge, and the dollar impact of doing so versus not.

Why it matters. Two practical implications, giving a forward-looking, market-implied framework for pricing and hedging LP loss:

- IL is a priced, tradeable **short-gamma** exposure: for full-range and sufficiently fee-dense concentrated positions, fee income can **meaningfully defray — and sometimes finance** — the hedge, turning a variable loss into a fixed, hedgeable cost.
- The band-edge cap is **asymmetric** (actionable for product design): a symmetric 70–130% range locks a *deeper* downside loss (–10.7%) than upside (–6.1%).

1. The unhedged LP loses — re-measured at the position level

The Topaz Blue cohort (17 pools, 43% of TVL) earned $199.3m in fees but suffered $260.1m of IL, a net $60.8m below HODL, leaving 49.5% of LPs with negative returns. We reproduce the direction of that finding independently and at the position level. Reconstructing every position's deposits and withdrawals from NonfungiblePositionManager *IncreaseLiquidity*/*DecreaseLiquidity* events keyed by token-ID, restricting to single deposit–withdraw cycles held at least 1 day (excluding intra-block 'flash' LPs, the only group the original study found did not lose), and valuing each position's terminal basket at its exact exit-minute price, we measure the realised IL of 27,529 closed positions across 5 flagship ETH pools. Every pool is negative; the total is –$50.7m (Fig1). **This section measures IL in isolation** — fee income is introduced separately in Section 3, where the net LP outcome is evaluated; 'every pool lost IL' is not the same claim as 'every LP lost money net of fees'.

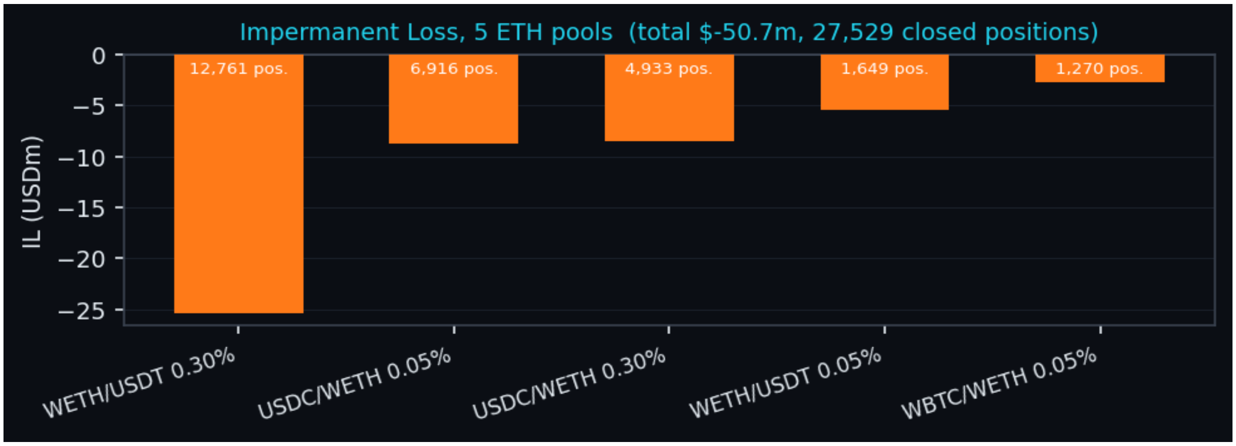

Fig1. Position-level measured IL by pool (Dune; trailing year). Counts above each bar are closed positions. These are **single-cycle, ≥1-day, closed** positions — a clean subset for **directional validation**, not a pool-wide IL total (see Limitations, end of §3).

2 estimator points are new relative to the original work. (i) **Pool-level Mint/Burn events cannot isolate positions**: the position owner is the NFT manager contract, collapsing thousands of users at 1 tick range into 1 blob; token-ID reconstruction is required. (ii) Because *IL = (W – D)·price* is **linear in price**, it equals the true loss only **at the exit price**; valuing at a daily average flips signs on high-churn pools. Exit-minute pricing resolved the last positive outlier and yielded uniformly negative IL.

Pool	Closed pos.	Measured IL ($m)	Avg loss / pos. ($)	% of total IL
WETH/USDT 0.30%	12,761	-25.4	1,987	50%
USDC/WETH 0.05%	6,916	-8.8	1,270	17%
USDC/WETH 0.30%	4,933	-8.5	1,731	17%
WETH/USDT 0.05%	1,649	-5.4	3,263	11%
WBTC/WETH 0.05%	1,270	-2.7	2,102	5%
All 5 pools	**27,529**	**-50.7**	**1,842**	**100%**

Table 1. Per-pool breakdown. The loss is **not concentrated in one pool**: per-position IL is comparable across pools ($1.3k–$3.3k), and although WETH/USDT 0.30% is ~half the dollar total, that reflects its far larger *position count*, not outsized individual positions. A holding-behaviour check (Q7) reinforces this: by pair, the median closed position is $2,598–$5,926 (retail-scale, not whales) and the average holding period is ~18 days (multi-week, not flash). We do not report a top-tail statistic, so a few large losers *within* a pool cannot be fully ruled out; the comparable per-position and median figures are consistent with a wide base rather than a few whales. Pool-wide annual fees (Q2/Q4) are used in Section 3, not netted here, to avoid conflating IL with net P&L.

2. IL is short gamma: pricing the hedge off the smile

An LP is short convexity‡: the pool sells the appreciating asset into a rally, ending with less of the winner — a concave payoff. The protection an LP would buy is the mirror image, a **long** options position. By the Carr–Madan identity, any twice-differentiable payoff is a static strip of options: $E^Q[g] = g(F) + \int g''(K)\cdot O(K)\, dK$, with O the out-of-the-money put (K<F) or call (K>F) — the undiscounted, forward-measure option value (we ignore discounting over short crypto tenors). For full-range (V2) IL the leading-order premium is the variance-swap value $\sigma^2 T/8$; for concentrated (V3R) IL the same strip is amplified inside the band by capital efficiency. We evaluate the integral under the SABR-fitted Deribit density (Hagan et al., 2002), sanity-checking the resulting V2/V3R premiums against indicative quote levels under comparable assumptions.

‡ Throughout, *short gamma* refers to the LP's payoff versus HODL as a function of terminal price (the comparison basis of the original study); this differs from the inventory-rebalancing sign conventions sometimes used in AMM discussions.

The capped 'Limited' (V3L) product is **not** a European iron condor: it **locks** the IL at the band edge on first touch, a path-dependent **first-passage barrier-style** payoff priced by first-passage Monte-Carlo. A static European strip prices only the terminal-clamp approximation; the gap to the barrier value is the path-dependence a vanilla strip cannot replicate — a distinction absent from the original report. A second, previously unremarked property is the **asymmetry of the band-edge cap** (Fig2): IL is symmetric under the price ratio $r \leftrightarrow 1/r$, not under linear distance, so a linearly symmetric band such as 70–130% caps a *deeper* loss on the downside (–10.7%) than the upside (–6.1%); the caps match only for a reciprocal-symmetric band (e.g. 70–143%).

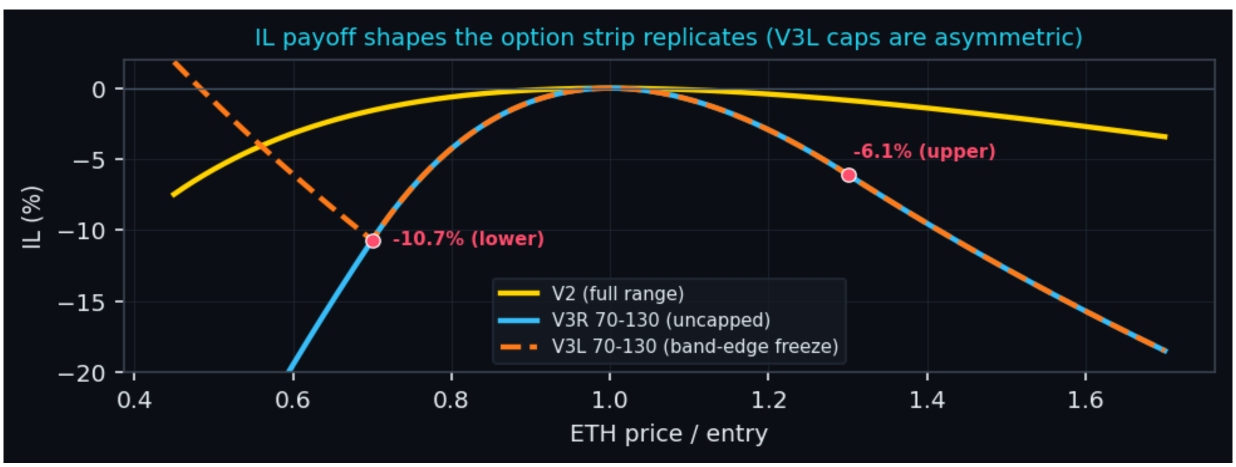

Fig2. IL payoff curves the option strip replicates; red points mark the asymmetric V3L first-touch caps.

Payoff glossary. **V2** — full-range LP; IL concave and uncapped on both sides. **V3R** (‘Regular’) — liquidity concentrated in a band [a,b]; IL amplified in-band by capital efficiency, uncapped beyond it. **V3L** (‘Limited’) — identical in-band, but IL **locks (freezes) at the band edge on first touch** — a path-dependent barrier, not a European condor.

Throughout, **notional** (equivalently, *position capital*) means the LP capital deployed at entry — not the virtual-liquidity notional inside the Uniswap range, which for concentrated positions differs. Surface: a **static, representative historical snapshot** of Deribit ETH options (SABR fit dated 2025-06-07); the premium grid below is an **illustrative model fair value** off that fixed snapshot, not a live quote — the method re-prices against any current surface, so the framework, not the snapshot date, is the contribution. (Deribit supplies the **implied-volatility surface** — option prices — not the spot feed used to measure IL in Section 1, which is Dune’s *prices.usd*.)

Indicative IL-protection premiums (model fair value, % of notional), priced off the smile:

Series	10d	30d	60d	90d
V2	0.10%	0.60%	0.86%	1.16%
V3R 50-150	0.43%	2.60%	3.60%	4.66%
V3L 50-150	0.43%	2.35%	2.88%	3.50%
V3R 70-130	0.72%	3.94%	5.17%	6.47%
V3L 70-130	0.72%	2.76%	3.23%	3.92%

These premiums are the price at which an LP converts a variable IL exposure into a **fixed, upfront cost**. Whether fee income covers that cost is concrete (Table 2): at the blended fee APR, 30-day fees are ~1.3% of capital, so a full-range hedge is comfortably covered while tightly concentrated hedges cost more per dollar of capital.

Series	30d hedge	30d fees*	cover (fees/hedge)	breakeven**
V2	0.60%	1.29%	2.1x	14 d
V3R 50-150	2.60%	1.29%	0.5x	61 d
V3L 50-150	2.35%	1.29%	0.5x	55 d
V3R 70-130	3.94%	1.29%	0.3x	92 d
V3L 70-130	2.76%	1.29%	0.5x	64 d

Table 2. Fee-cover and breakeven for the 30-day hedge, as % of position capital. *30d fee income at the blended 15.7% APR (flat, capital-efficiency-agnostic). **Holding days at which cumulative fees equal the premium (premium ÷ fee APR). Full-range (V2) protection is comfortably fee-covered (~2x; breakeven ~14 days); tightly concentrated hedges cost more at a flat APR, but concentration earns a *higher* fee APR (capital efficiency) that scales both the fee numerator and the premium, which could pull the V3 ratios back toward V2’s, depending on realised fee density.

In hedge terms: **for the LP** — buy IL protection, keep the fees, retain the residual upside; **for the protection seller** — collect the premium for warehousing the LP’s short-gamma risk. The product thesis is that, for full-range and moderately concentrated positions, fee income meaningfully defrays the cost of hedging IL exposure.

3. The rebalancing gamma bleed, quantified

Topaz Blue and Bancor explicitly compared rebalancing to **delta-hedging gamma bleed** — once rebalanced the loss is no longer impermanent — and left the Monte-Carlo analysis to future research†; this section supplies it. We calibrate to on-chain realised vol and fee/TVL data, with the path *shape* informed by the fitted options skew: 90-day realised ETH volatility of 41% (Dune), a TVL-weighted blended fee APR of 15.7% across the 5 pools, and the fitted SABR skew. Fig3 reports mean net return versus HODL and the share of paths finishing below HODL, for passive-hold versus rebalance-on-exit.

† No published sequel followed from the original authors: Bancor sunset its on-chain IL-protection in June 2022, and the thread it flagged was advanced not by Topaz Blue / Bancor but by the loss-versus-rebalancing (LVR) literature (Milionis et al., 2022).

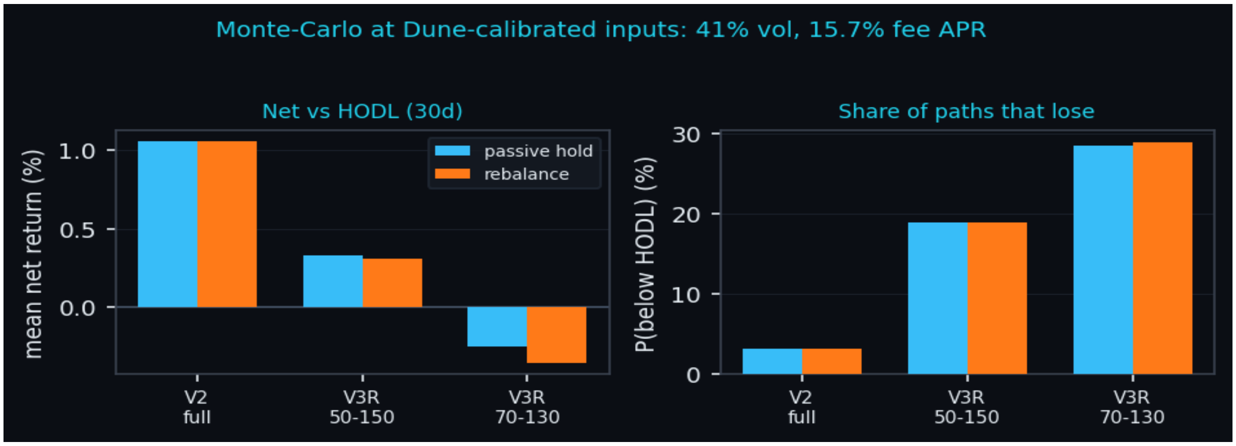

Fig3. Calibrated Monte-Carlo (30-day horizon). Concentration sharply raises the loss probability; the hold-vs-rebalance gap is small at this horizon (see text), and V2 full-range never exits its band, so its 2 bars are identical by construction.

Model details. Each scenario simulates **40,000** price paths over a **30-day** horizon in **90** time-steps (≈8-hour steps) under the lognormal β=1 SABR SDE ($dF=\alpha F\, dW_1$, $d\alpha=\nu\alpha\, dW_2$, corr ρ) with the fitted ρ and ν and a **drift of zero** (we take no directional view; vol level is the Dune realised figure). Fees accrue at the pool APR while price is **in range**. The rebalance rule, on a band exit, **re-centres** the band at spot, **crystallises** the IL accrued since entry, and pays a **15 bp** cost; V3L instead freezes IL at the edge on first touch. Net return per path = fees + IL − costs; P(loss) is the share of paths with net < 0. Results are stable to the seed at this path count; an accompanying implementation dashboard exposes all of these as adjustable inputs.

At these inputs the wide full-range position is mildly profitable (mean net +1.1%, P(loss) 3%), but a tightly concentrated position turns negative (V3R 70–130 mean net -0.3%, P(loss) 28%), and rebalancing-on-exit makes it worse still — the bleed the original study flagged. The hold-vs-rebalance bars look **nearly identical** because, at a 30-day horizon, 2 effects roughly cancel: re-centring keeps fees flowing (a *passive* position out of range earns nothing), while each band exit **crystallises** the IL accrued since entry and pays a ~15 bp cost. The bleed is therefore real but second-order here; it **compounds** with horizon, volatility and the number of band exits, so over longer holds or tighter, more-frequently-breached bands the gap widens materially (Fig4; V2 full-range never exits, so its 2 bars coincide exactly).

Fig4 makes the compounding explicit. The mean drag (hold net minus rebalance net) rises from ~**10 bp at 30 days to ~400 bp by 90 days** (70–130 band), and from a few bp for a wide 50–150 band to ~**200 bp** at 92–108 (30-day) — the second-order bleed of Fig3 becoming first-order as exits multiply.

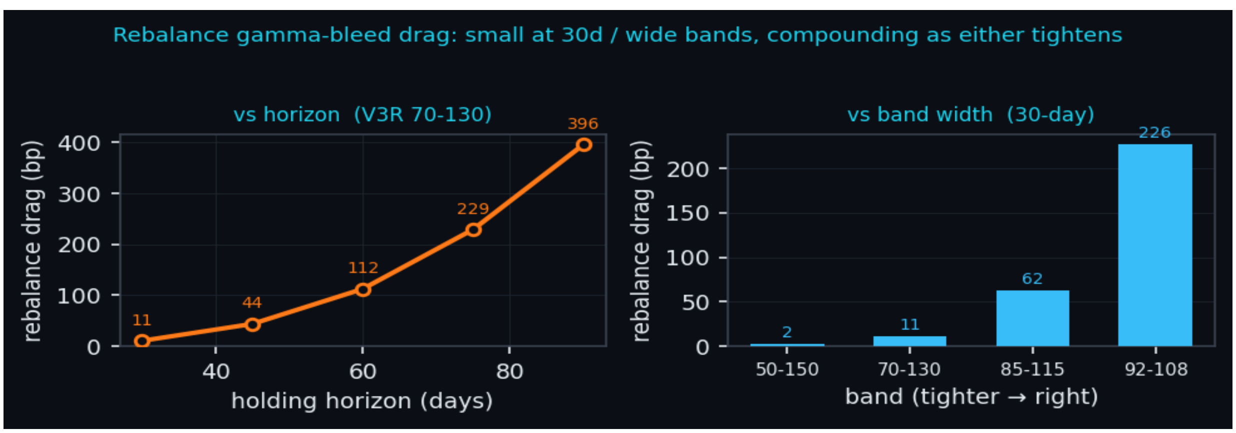

Fig4. Rebalance drag (mean passive-hold net – mean rebalance net, bp) under the same calibration. Left: vs holding horizon (V3R 70–130), steps scaled to hold the ~8h exit granularity constant. Right: vs band width at a 30-day horizon. The drag is an expected effect, not a per-path guarantee.

The practical implication closes the loop with Sections 1–2: because the loss is a priced, tradeable short-gamma exposure, a fairly-valued IL hedge purchased off the smile can move the LP's expected outcome from negative back to positive, partly or fully financed by fee income.

Limitations. Beyond the sampling caveat at Fig 1, the measured IL is priced at Dune's *prices.usd* minute-level feed, matched at each position's exit minute (an external aggregated price), rather than the pool's own tick price, and, as the original authors stress, IL in USD is not invariant under a change of numeraire when capital is added and withdrawn over time; the single-cycle, closed-position filter may also underweight professional, actively-managed (multi-cycle) LPs, so the sample is clean but behaviourally selected. Separately, because first-touch (V3L) protection is **path-dependent**, its price is **model-dependent even when the terminal smile is matched** — the barrier value is not pinned down by the vanilla smile alone. These constrain the dollar-total interpretation, not the directional validation.

4. Worked examples: what hedging would have done

Sections 1–3 work at the pool level. Here are 3 **representative positions** from the measured cohort — each a real pool with the cohort-**median** size, **average** hold (Q7) and **measured fee APR** — one of each payoff type: a **full-range (V2-style)** position, a concentrated uncapped **V3R**, and a capped 'Limited' **V3L** (all are v3 positions; 'V2' denotes the full-range payoff, not a literal Uniswap v2 pool). Each LP follows the same 4 steps — *enter, price the hedge off the Section-2 grid, hold, settle* — and for a replicated hedge the outcome collapses to a single, **move-independent** number: *net = fees – premium*.

Ex. A — USDC/WETH 0.05%, full-range (V2-style). Deploy **$2,600** for ~**40 days**; the price moves **+35%** (2.6σ over 40d at 41% vol). The matching hedge is priced at **0.7%** = $18 upfront; fees accrue **+$45** and realised IL is **–$29** (-1.1%). Unhedged net **+$16**; hedged net **+$27** (fees – premium). Full-range protection is cheap and comfortably fee-covered — the hedge removes all IL risk for a small fixed cost.

Ex. B — WETH/USDT 0.30%, V3R 70-130 (7.0× capital efficiency). Deploy **$3,000** for ~**18 days**; the price moves **-35%** (3.9σ over 18d at 41% vol), breaching the band where the **uncapped** loss keeps deepening. The matching hedge is priced at **2.0%** = $60 upfront; fees accrue **+$31** and realised IL is **–$449** (-15.0%). Unhedged net **–$419**; hedged net **–$29** (fees – premium). Concentration amplifies the loss and, being **uncapped**, V3R keeps bleeding as the price falls; the hedge cuts a **–$419** result to **–$29**, saving **$389**.

Ex. C — WBTC/WETH 0.05%, V3L 50-150 (4.2× capital efficiency). Deploy **$5,926** for ~**60 days**; the price moves **-55%** (3.3σ over 60d at 41% vol), touching the lower edge so IL **freezes at the -22.3% first-touch cap**. The matching hedge is priced at **2.9%** = $171 upfront; fees accrue **+$60** and realised IL is **–$1,322** (-22.3%). Unhedged net **–$1,262**; hedged net **–$111** (fees – premium). The 'Limited' cap freezes IL at the edge, yet in a deep crash with thin fees it is still a large dollar loss; the **first-touch barrier hedge** (the value a European strip only approximates, §2) cuts **–$1,262** to **–$111** — saving **$1,151**.

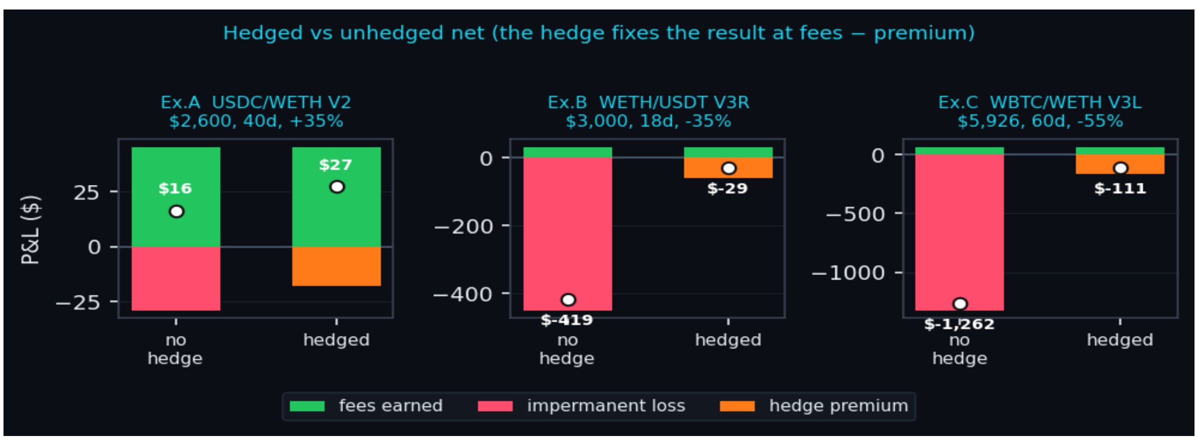

Fig5. Per-example P&L. Each 'no hedge' bar nets fees (green) against IL (red); each 'hedged' bar nets the same fees against the upfront premium (orange), since the hedge payout cancels the IL. White dot = net P&L. Panels use separate y-axis scales. Representative positions, not individual on-chain rows.

The pattern mirrors the framework: hedging never promises a profit — it **converts a variable, open-ended IL into a fixed, pre-known cost**, and earns its keep precisely where Section 3's bleed is largest: concentrated bands, capped products, and sharp moves.

5. Future work: Impermanent Gain and the hedged-LP book

This paper prices the loss; the sequel manages it — modelling **Impermanent Gain (IG)**, the long-gamma mirror an LP buys to hedge, via the Black–Scholes-tailored LP Greeks of Bardoscia & Nodari (2023). An unlocked LP position is **+Delta, –Gamma, +Theta, ~0 Vega**; the IG instrument is its mirror, and the hedged book turns net-long volatility. 4 threads:

- **The hedged LP's net P&L** — combine the LP with its IG hedge into a +Vega, reduced-gamma book, and extend the Table-2 fee-cover into a full net-P&L distribution: when is it positive?

- **The protection seller's economics** — *premium – E[IG paid]*, the volatility risk premium for warehousing LP short-gamma, across over- and under-priced-vol regimes.
- **Reconciling 2 pricing lenses** — the BS-tailored IG premium (LP Greeks) versus the Carr–Madan smile-replicated premium of Section 2; under matched assumptions they should agree.
- **Fee-dominant regimes & optimal band selection** — when realised fees beat realised IL, and which band / Regular-vs-Limited choice maximises expected fee-cover, using the asymmetric-cap result of Section 2.

Appendix A. On-chain methodology (Dune Analytics)

Every input was computed from public Ethereum data on **Dune Analytics**. The position-level IL estimator required 8 iterations (Table A1): each intermediate failure was diagnostic, and the decisive fix was valuing each position at its *exact exit-minute* price.

Table A1. 8-step derivation of the position-level IL estimator.

Step	Estimator	Outcome	Lesson
1	aggregate Mint - Burn, valued at close	mixed (1 gain)	net flow x price is not IL
2	per-position match by owner + tick range	3/5 negative	owner = NFT manager collapses positions
3	same match, valued at exit-day price	4/5 positive	owner+tick matching is broken - need token-ID
4	NFPM token-ID, exit-day price	3/5 negative	right source; multi-cycle positions leak
5	single deposit->withdraw cycle only	4/5 negative	JIT / flash pool still positive
6	+ hold >= 1 day (exclude JIT)	4/5 negative	not only JIT - deeper cause remained
7	all-time deposit capture	unchanged	window truncation ruled out
8	value at exact exit-MINUTE price	5/5 negative	IL identity holds only at the exit price

Appendix B. Dune query map

Each Q-code in the text maps to a named Dune query: **Q1** pool-level Mint/Burn aggregate (baseline proxy); **Q2** fee tier + trailing 30-day fees → annualised fee APR (Section 3); **Q3** pool fees vs IL proxy cross-check; **Q4** pool TVL / capital base (Section 3, Table 1); **Q5** per-position IL by owner + tick range (superseded); **Q6(a–e)** NonfungiblePositionManager token-ID per-position IL, single-cycle & ≥1-day iterations (Fig 1, Table 1); **Q7** holding time + median position size by pair (Table 1); **Q8** final estimator with exact exit-minute pricing.

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